
Weekly Report: [W06]

General Information

- **Group ID:** 52
- **Group Name:** Group 52
- **Class:** 25A01
- **Project Name:** Super Mario Game
- **Date range:** 2026-07-12 – 2026-07-18
- **Github Repository:** <https://github.com/ndmhuy/SuperMarioGame.git>

Tasks Completed This Week

25125083 - Nguyễn Đình Minh Huy (Member A - Engine, Physics & Player/Item Entities)

Branch: `A/feature/save-load` (*Save/Load, Persistence, HUD settings*)

- **Task 1: Save/Load Slots Serialization:** Created `Serializer` managing JSON slots schemas (character type, polymorphic player state, scores, star coins arrays, current checkpoint positions, play time). Safely creates target save directories if missing.
- **Task 2: Gameplay Statistics Tracker:** Implemented `StatisticsTracker` subscribing to the EventBus (tracks collected coins, defeated enemies, deaths, max combo achievements, and cumulative elapsed playtime).
- **Task 3: Achievement Notifier HUD:** Implemented `AchievementManager` condition check routines for all 10 core achievements. Designed a slide-in/slide-out toast HUD system that announces unlocked achievements.
- **Task 4: Interactive Pause & Settings Panel:** Created an ImGui Pause HUD panel wiring slots preview headers, manual save triggers, volume controls, keybindings display, difficulty sliders, and checkpoint auto-saves.
- **Task 5: Integration Tests Suite:** Coded `verify_save_load.cpp` executing slot operations and achievements conditions checks.

Branch: `A/feature/level-editor` (Mario Maker Level Editor)

- **Task 1: Undo/Redo Commands Framework:** Implemented the Command Pattern in `EditorCommands` wrapping grid placements/deletions for full command history rollbacks.
 - **Task 2: Interactive Palettes HUD:** Developed a `MapEditor` viewport dashboard featuring tile brush combos, entity brush arrays, action triggers (Undo/Redo), and file save inputs.
 - **Task 3: Coordinates Mapping & Grid Layout:** Implemented pixel-to-coords viewport translation inside `Game::getMouseWorldPosition` to map mouse coordinates safely without exposing `sf::RenderWindow` reference. Implemented soft grid line rendering.
 - **Task 4: DRY Code Centralization:** Centralized duplicate tile and entity string name mappers from `LevelLoader` and `MapEditor` into a unified `SerializationUtils`.
 - **Task 5: Map Editor Test Harness:** Created `verify_map_editor.cpp` verifying grid updates, undo commands, and serialization.
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25125084 - Trần Gia Huy (Member B - Input, Sound & Gameplay Systems)

- **Task 1: Weekly Progress Sync:** Participated in architectural reviews and resolved cross-branch merge conflicts. Refined sound effects triggers inside the Achievement System integrations.
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AI Usage Declaration

- Used AI to structure the Command Pattern stack rollback allocations for entity replacements.
 - Used AI to design horizontal run-length serialization optimizations inside the Level Loader JSON output.
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Tasks Planned for Next Week

- **Nguyễn Đình Minh Huy:** Merge `A/feature/save-load` and `A/feature/level-editor` task branches to `dev`. Begin Phase 7 (Game States, Main Menu UI, World Map selection layers) and integrate Time Rewind circular buffer mechanic.
 - **Trần Gia Huy:** Complete Phase 5.3 HUD minimap and begin Phase 10 console debug overlays.
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Issues & Resolutions

Issue 1: Coin Collected Increment Bug in Achievement Observer

- **Problem:** When a player collected multiple coins simultaneously (such as a coin block dispensing 10 coins), the AchievementManager only registered a single increment of `m_coinsThisRun` because the EventBus observer was doing `m_coinsThisRun++` instead of reading the event data payload.
- **Resolution:** Modified the `CoinCollected` switch-case inside `AchievementManager::handleEvent()` to attempt an `any_cast<int>` on the event data, correctly accumulating the amount payload.

Issue 2: Aggregate sf::Vertex Initialization Compiler Error

- **Problem:** Compiling with SFML 3.0.2 threw errors when instantiating `sf::Vertex` using parenthesis syntax (`sf::Vertex(...)`) due to the removal of overloaded constructors for aggregate types.
- **Resolution:** Replaced the constructors calls with brace bracket initialization syntax (`sf::Vertex{...}`) which satisfies aggregate type requirements for both SFML 2.x and 3.x.